AMRITA VISHWA VIDYAPEETHAM | School of Engineering

Embedded Systems Project

Ghost Vehicle in Fog Detector Using Pressure Wave Sensing Before Headlights Appear

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INTRODUCTION

Seeing the unseen: Predicting oncoming traffic before the headlights cut the fog

The Invisible Threat

Dense fog critically reduces visual range on highways and rural roads, rendering standard headlights ineffective.

Drivers face severely delayed detection of approaching vehicles, drastically increasing the likelihood of high-speed, catastrophic collisions.

The Edge-AI Solution

A New Paradigm: Detecting approaching vehicles before they become visible by capturing the low-frequency pressure waves they push ahead of them.

The Technology Stack: Utilising highly sensitive MEMS acoustic sensors, the system processes physical data locally on a Cortex-M microcontroller.

Zero-Latency Intelligence: By deploying TinyML and Edge-AI methodologies directly on the chip, we trigger life-saving early warnings without relying on cloud processing or visual line-of-sight.

SENSE: Capture invisible pressure disturbances generated by moving vehicles.

PROCESS: Execute real-time signal analysis natively on a Cortex-M core.

WARN: Trigger immediate visual (LEDS/Display) and audible (Buzzer) driver alerts.

LITERATURE SURVEY

Study of the research work

S.NO	Title of the Paper	Year	Objective	Hardware /Software	Outcome	Inference
1.	**Adaptive Sensor Fusion for Robust Perception in Dense Fog: A Gated Vision and LiDAR Integration Framework** hhttps://www.mdpi.com/1424-8220/26/12/3728	JUNE 2026	Improve obstacle detection in dense fog using adaptive multi-modal sensor fusion.	**Hardware:** Gated Camera, 3D LiDAR. **Software:** AFW-Net, Attention-based Adaptive Fusion, Uncertainty Estimation, Cross-modal Feature Extraction.	Achieved 82% AP at 50 m visibility, improving detection by 23.7% over conventional RGB-LiDAR fusion.	Adaptive sensor weighting, Reliable sensor fusion, Robust fog perception
2.	**Ensemble Artificial Intelligence Framework for Real-Time Automatic Vehicle Detection and Classifications** https://ieeexplore.ieee.org/abstract/document/11363414	May 2026	Develop an ensemble AI framework for accurate vehicle detection and classification under adverse weather.	**Hardware**: RGB Camera. **Software**: YOLOv5, YOLOv7, YOLOv8, YOLOv10, DETR-ResNet50, Faster R-CNN, Dynamic Weighting, NMS, Ensemble Learning.	Achieved 96.6% mAP, 94.2% Precision, and 94.1% F1-score for multi-class vehicle detection.	Ensemble learning, High detection accuracy, Vision-dependent
3.	**FogGate-YOLO: Traffic Object Detection in Foggy Environments Using Channel Selection Mechanisms** https://www.mdpi.com/1424-8220/26/6/1811	MAR 2026	Design a lightweight object detector specifically optimized for foggy environments.	**Hardware:** RGB Camera. **Software:** YOLOv12, A2C2f-FSA Module, Bidirectional Feature Fusion Module (BFFM), GSConv, F-WIoU v3 Loss.	Improved mAP@0.5 by 5.2% (RTTS) and 7.3% (VOC-FOG) while maintaining real-time inference.	Fog-aware feature enhancement, Real-time detection, Camera limitations
4.	**FogFusion: Robust 3D object detection based on camera-LiDAR fusion for autonomous driving in foggy weather conditions** https://journals.sagepub.com/doi/abs/10.1177/09544070251327229	MAR 2026	Improve 3D object detection in fog using enhanced camera-LiDAR fusion.	**Hardware:** RGB Camera, LiDAR. **Software:** Depth Completion Network with Fog Convolution (DCFC), Flexible Cylindrical Voxel (FCV), Cylindrical Fusion Module (CFM).	mproved detection performance by ≥3.32% over baseline methods on KITTI-C.	Enhanced 3D perception, Virtual point clouds, Costly sensor fusion
5.	**Dynamic Visibility Recognition and Driving Risk Assessment Under Rain–Fog Conditions Using Monocular Surveillance Imagery** hhttps://www.mdpi.com/2071-1050/18/2/625	JAN 2026	Estimate road visibility and driving risk under rain-fog conditions using roadside surveillance images.	**Hardware:** Monocular CCTV Camera. **Software:** Koschmieder Scattering Model, Dark Channel Prior, Monte Carlo Simulation, Low Visibility Risk Field Force (LVRFF).	Achieved >86% visibility estimation accuracy and effectively identified high-risk driving zones.	Visibility estimation, Risk assessment, Infrastructure-based monitoring

LITERATURE SURVEY

Study of the research work

S.NO	Title of the Paper	Year	Objective	Hardware /Software	Outcome	Inference
6.	**ECE-VDTDA: A Robust and Computationally Efficient Collision Avoidance System for Driver Assistance in Foggy Weather**	**FEB 2026**	Enhance performance in foggy-weather using a cost-effective vehicle detection system for distance, speed, and TTC-based collision warnings.	**Software:** SimYOLO-V5s_WIOU detector, Deep-SORT-based tracking **Hardware:** RGB cameras with edge-capable platforms for real-time inference	SimYOLO-V5s_WIOU improved fog detection accuracy (up to +17.45% mAP50) on foggy driving, enhanced tracking reliability, and enabled TTC-based graded collision warnings.	SimSPPF and Wise-IoU improve fog detection reliability, while robust tracking and TTC-based warnings support safer collision avoidance.
7.	**Multi-Target Tracking with Collaborative Roadside Units Under Foggy Conditions**	**JAN 2026**	Improve foggy roadside LiDAR tracking by reducing point-cloud noise and enabling collaborative multi-RSU perception.	**Hardware:** Roadside LiDAR-equipped RSUs for fog-affected traffic data collection. **Software:** Fog-noise removal using adaptive filtering and multi-RSU cooperative tracking with PHD filtering	Improved detection accuracy by +8% (thin fog) and +29% (thick fog), with enhanced multi-class tracking performance.	Noise-aware denoising and multi-RSU PHD fusion improve foggy LiDAR detection, range, and tracking stability.
8.	**Fog-YOLO: A Real-Time Object Detector for Foggy Autonomous Driving Scenarios**	**FEB 2026**	Develop a low-latency, fog-robust detector for autonomous driving with improved accuracy.	**Hardware:** RGB cameras with GPU-enabled platforms for real-time inference. **Software:** Fog-YOLO with fog-aware modules and improved detection head	Improved mAP over YOLO baseline with better small/occluded object detection in varying fog while maintaining real-time performance.	Fog-aware features and improved detection heads enhance bounding boxes and reliability for foggy perception.
9.	**3D Vision Object Detection for Autonomous Driving in Fog Using LiDAR**	**APR 2025**	Improve foggy 3D object detection using LiDAR-based perception and fog-aware processing.	**Hardware:** Automotive LiDAR sensors evaluated under foggy conditions. **Software:** LiDAR-based 3D detection with fog-mitigation techniques	Improved 3D detection accuracy and recall under fog compared to uncompensated LiDAR baselines.	Fog-aware processing preserves point-cloud features, enabling reliable 3D detection in poor visibility.
10.	**Simulation of LiDAR Under Fog with Generative Adversarial Networks for Robust 3D Object Detection**	**OCT 2025**	Generate realistic foggy LiDAR data using WGAN–LSTM to improve 3D detector robustness without real fog data collection.	**Hardware:** Velodyne VLP-32C LiDAR with fog-chamber and onboard ROS **Software:** ROS-based LiDAR processing with WGAN-GP + LSTM fog simulation	Generated 80,000 foggy LiDAR frames using WGAN–LSTM, improving 3D detector robustness and reducing the need for real fog data collection.	GAN-based fog simulation reduces real-world data dependence and improves autonomous vehicle perception robustness.

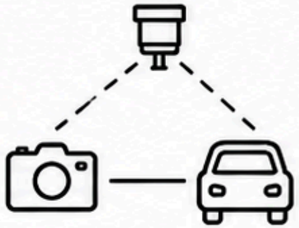
INFERENCE FROM LITERATURE SURVEY

1. Existing Fog-Based Vehicle Detection Approaches



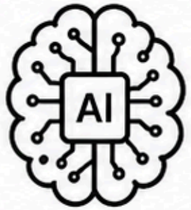
Camera, LiDAR and AI-based detection in fog

2. Sensor Fusion-Based Detection Systems



Gated Camera + LiDAR improves accuracy but high cost & complex

3. AI & Deep Learning Based Methods



YOLO, Faster R-CNN, FogGate-YOLO give high accuracy but depend on visual information

4. Limitations of Existing Systems



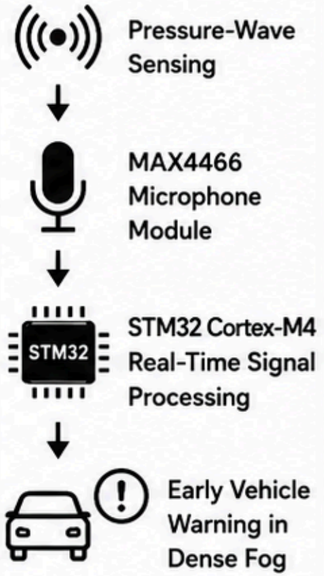
- Complex image & point cloud processing
- Focus on visibility assessment
- Cannot detect vehicles before visual identification

5. Research Gap Identification



- Need low-cost, lightweight, real-time system
- Early detection of hidden vehicles before visual confirmation

6. Proposed Solution

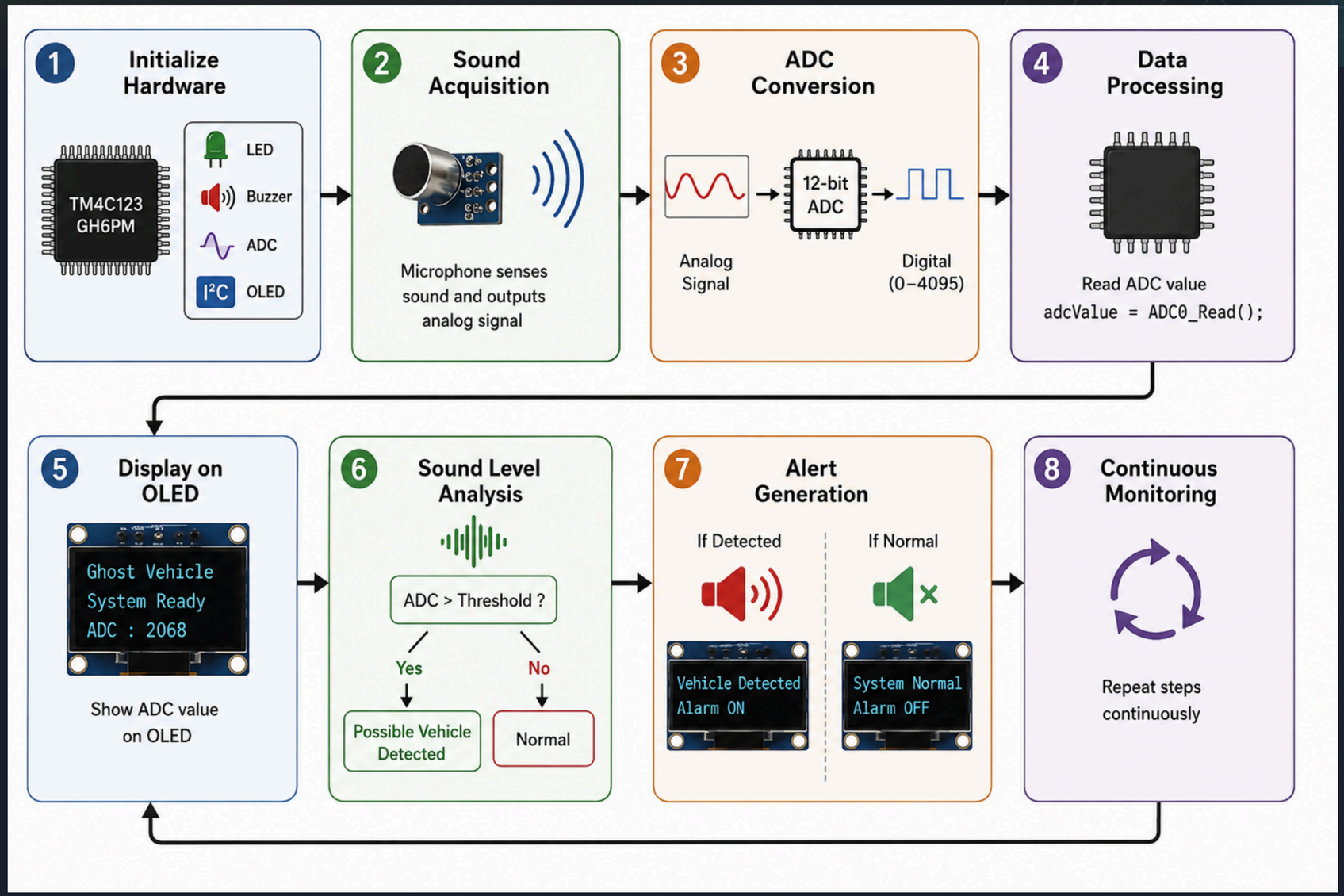


12-Week Development Sprint

A structured roadmap from literature survey to functional Edge-AI prototype.

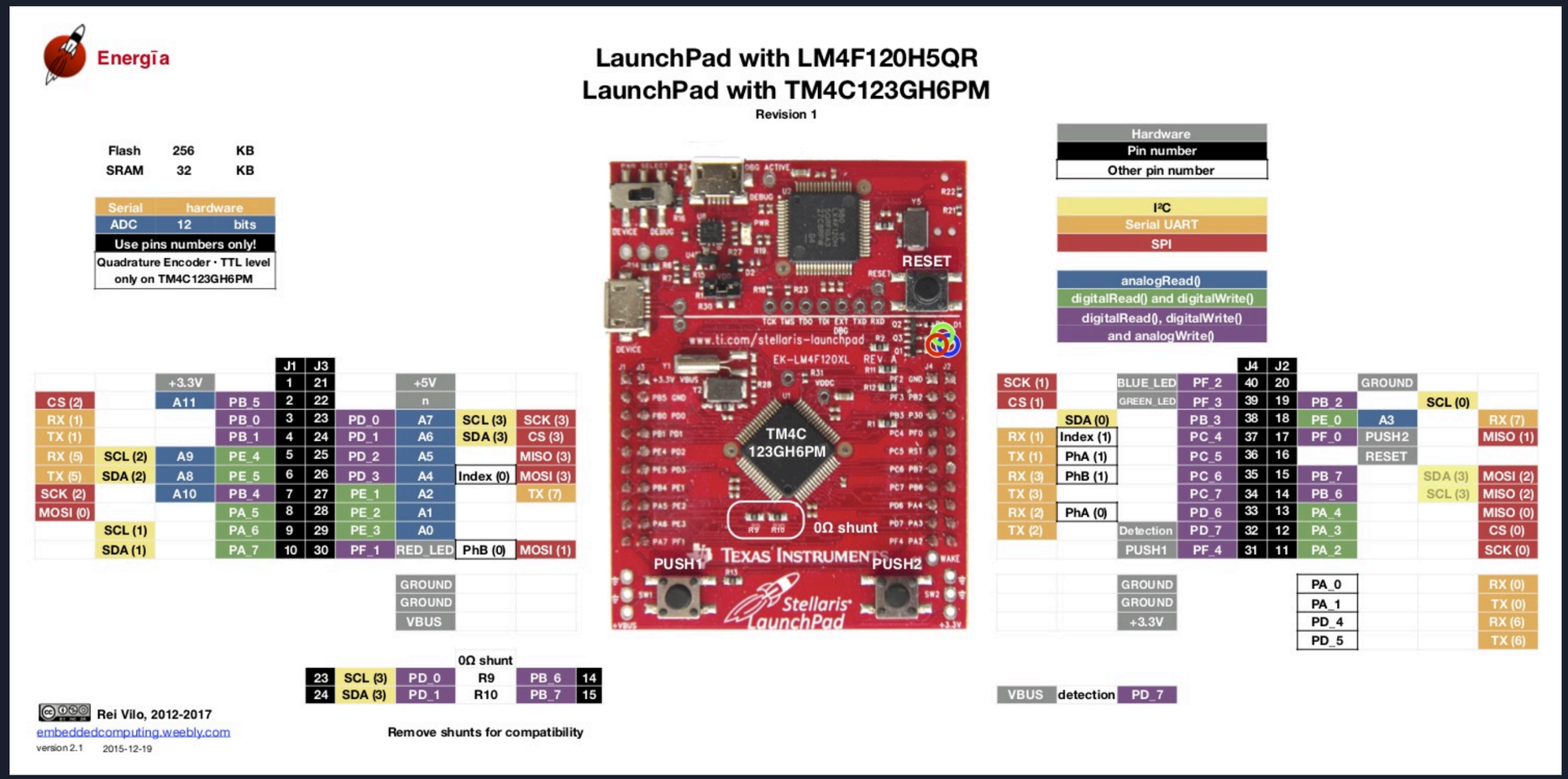
Phase & Focus	Week	Core Activity	Key Milestone
Phase 1: Foundation (Research & Hardware)	Week 1-2 Week 3	Literature Survey, Problem Analysis & Component Procurement. STM32/Cortex-M setup and architecture learning.	Hardware Procured & Environment Setup
Phase 2: Integration (Interfacing & Data)	Week 4 Week 5 Week 6	Sensor Interfacing & ADC Testing. Data Collection from live moving vehicles. Initial Signal Processing Implementation.	Successful Sensor Data Acquisition
Phase 3: Edge-AI (Algorithm & Alert)	Week 7 Week 8 Week 9	Vehicle Detection Algorithm (TinyML) Development. LED, Buzzer, and Display Integration. System Testing, Calibration, and Optimisation.	Functional Detection Algorithm
Phase 4: Finalisation (Prototype & Delivery)	Week 10 Week 11 Week 12	Complete Prototype Assembly. Result Analysis and Performance Evaluation. Report Writing and PPT Preparation.	Final Demonstration (Oct 6)

methodology



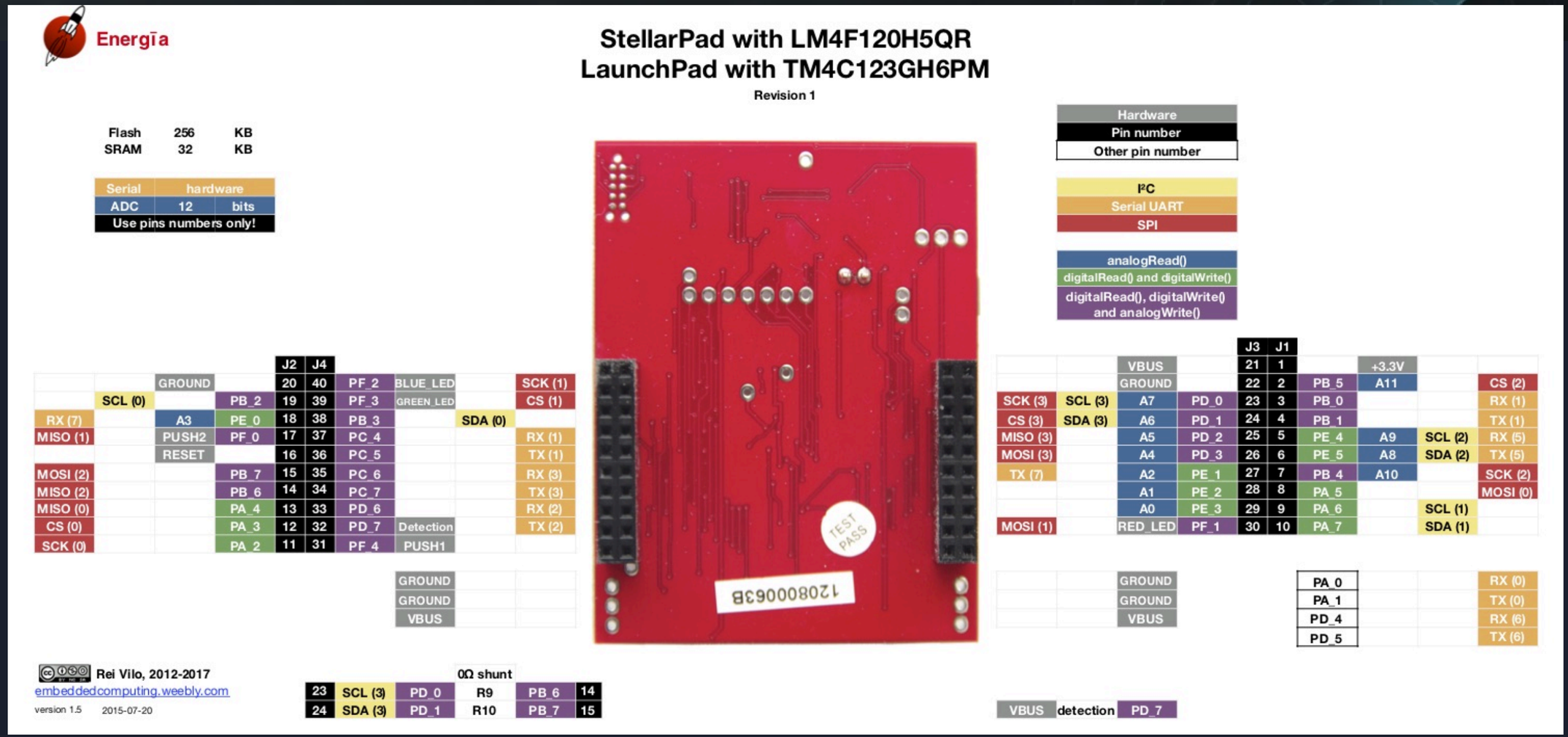
HARDWARE COMPONENTS

The TM4C123GH6PM is the central microcontroller of our project that acquires microphone signals, processes vehicle detection logic, and controls the OLED display, buzzer, and LEDs to provide real-time collision warnings in foggy conditions.



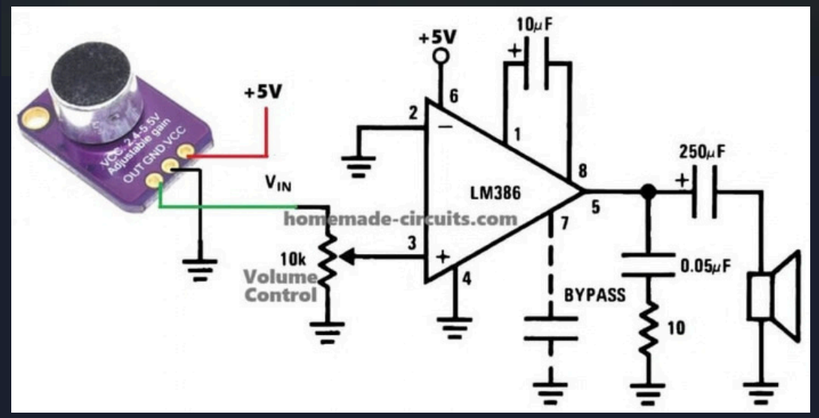
TM4C123GH6PM

HARDWARE COMPONENTS



TM4C123GH6PM

HARDWARE COMPONENTS



MAX4466 Analog Microphone Module



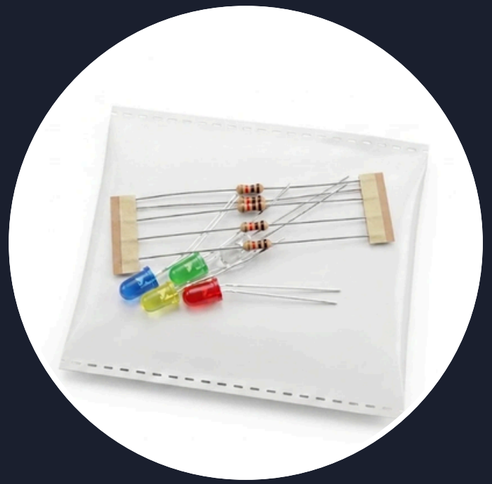
ST Link V2 Programmer



BUZZER



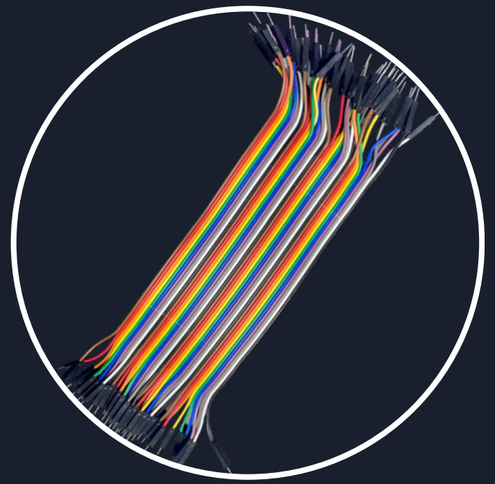
SSD1306 OLED Display



LEDS AND RESISTORS



BREADBOARD

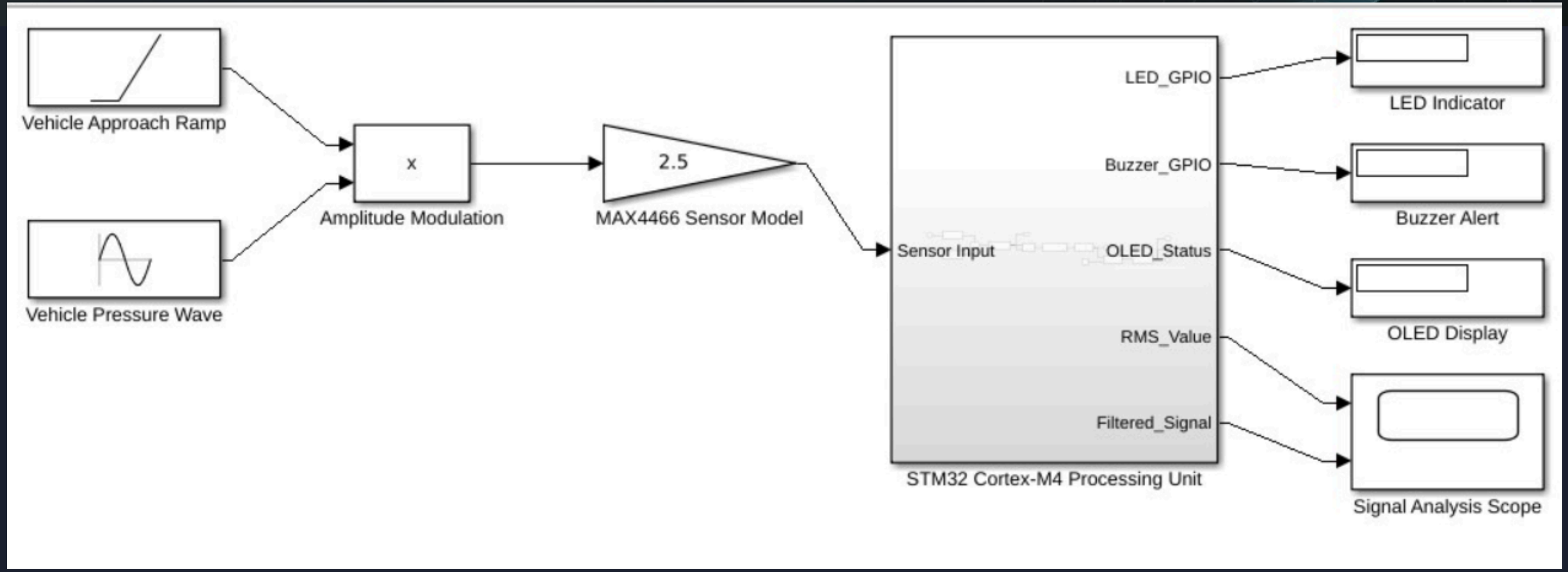


JUMPER WIRES



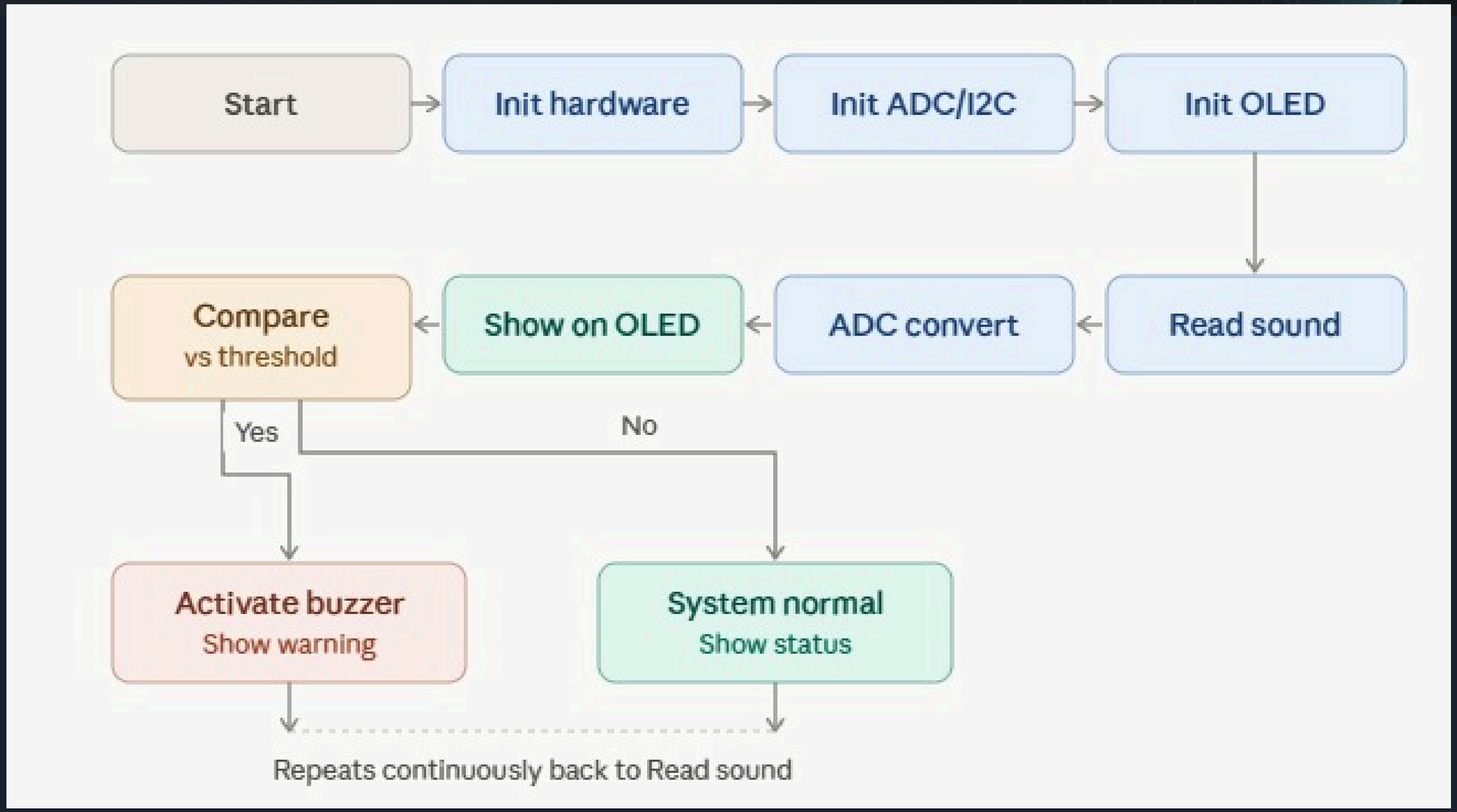
USB CABLE

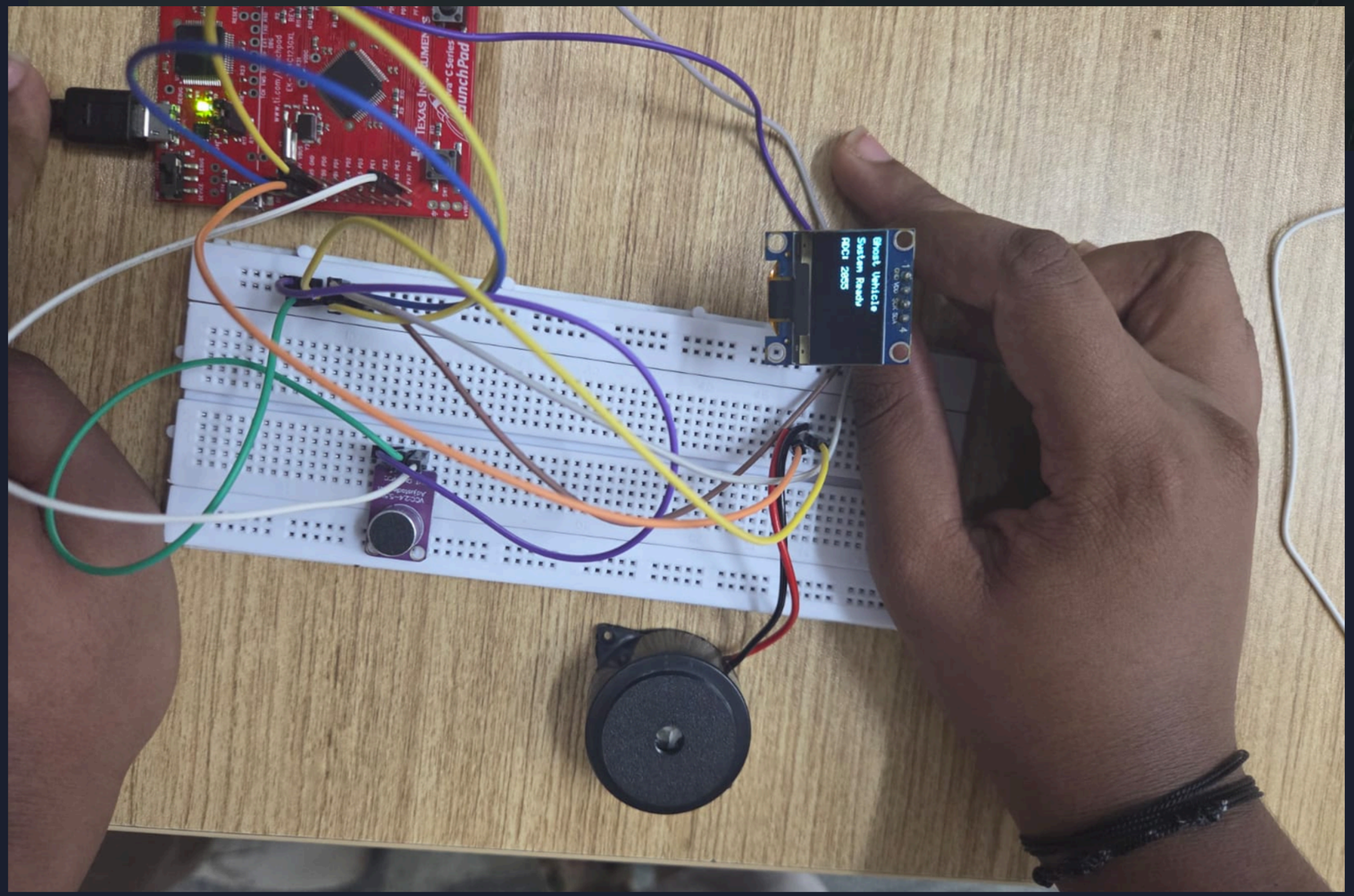
DEVELOP SIMULATION MODEL



The Simulink model represents the proposed ghost vehicle detection system, where vehicle-generated pressure waves are simulated and processed through the MAX4466 sensor model. The STM32 Cortex-M4 processing unit performs signal analysis and generates real-time alerts through LED, buzzer, and OLED display when a vehicle is detected

BLOCK DIAGRAM





HARDWARE MODEL

RESULTS

1. Successfully interfaced the **MAX4466 microphone sensor** with the TM4C123GH6PM microcontroller.
2. Real-time ADC values were successfully acquired from the **pressure-wave sensor.**
3. The ADC reading was displayed correctly on the **OLED display via I²C communication.**
4. The display confirmed that the Ghost Vehicle Detection System was initialized and functioning properly.
5. The prototype successfully demonstrated real-time acoustic pressure-wave monitoring.
6. The obtained results validate the feasibility of the proposed sensing approach for future ghost vehicle detection.

REFERENCES

[1]. Zhang, Fengyuan, et al. "Adaptive sensor fusion for robust perception in dense fog: A gated vision and LiDAR integration framework." Sensors 26.12 (2026): 3728.

[2]. Q. Yuan et al., "Ensemble Artificial Intelligence Framework for Real-Time Automatic Vehicle Detection and Classifications," in IEEE Access, vol. 14, pp. 16034-16050, 2026, doi: 10.1109/ACCESS.2026.3657829.

keywords: {Vehicle detection;Artificial intelligence;Accuracy;Real-time systems;Detectors;YOLO;Autonomous vehicles;Automobiles;Robustness;Feature extraction;Deep learning;dynamic weighting;ensemble learning;real-time vehicle detection},

[3].Wang, Quanxiang, Zhaofa Zhou, and Zhili Zhang. "Fog-YOLO: A Real-Time Object Detector for Foggy Autonomous Driving Scenarios." Symmetry 18.2 (2026): 322.

[4]. Zhang, Bingli, et al. "FogFusion: Robust 3D object detection based on camera-LiDAR fusion for autonomous driving in foggy weather conditions." Proceedings of the Institution of Mechanical Engineers, Part D: Journal of Automobile Engineering 240.4 (2026): 2312-2322.

[5].Xie, Zilong, et al. "Dynamic Visibility Recognition and Driving Risk Assessment Under Rain–Fog Conditions Using Monocular Surveillance Imagery." Sustainability 18.2 (2026): 625.

Thank You!